
Title of Your Final Year Design Project

By

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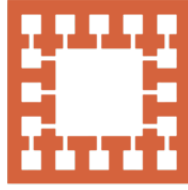
Submitted in partial fulfilment of the requirements
of the degree of Bachelor of Science in Computer Science and Engineering

July 11, 2026



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
VARENDRA UNIVERSITY

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DECLARATION

We hereby declare that the work presented in this thesis/project entitled "**Your FYDP Title**" is the result of our own research carried out under the supervision of "**Supervisor Name**".

We further declare that this thesis has not been submitted previously, in whole or in part, for any degree or diploma at any other university or institution. All referenced materials have been duly acknowledged.

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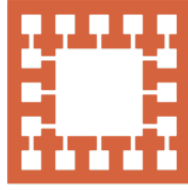
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CERTIFICATE

This is to certify that the thesis/project entitled "**Your FYDP Title**" has been submitted by **Student Name 1 (ID), Student Name 2 (ID), Student Name 3 (ID)** in partial fulfillment of the requirements for the degree of **Bachelor of Science in Computer Science and Engineering** at **Varendra University, Rajshahi, Bangladesh**.

The work presented in this thesis is an original contribution of the above-mentioned students and has been carried out under my direct supervision. To the best of my knowledge, this work has not been submitted, either in whole or in part, for any other degree or qualification at any other institution.

Supervisor

Supervisor Name

Designation

Department of Computer Science and Engineering

Varendra University

Acknowledgements

This work would have not been possible without the input and support of many people over the last two trimesters. We would like to express my gratitude to everyone who contributed to it in some way or other.

First, we would like to thank my academic advisors, ...

Our sincere gratitude goes to ...

We are also thankful to ...

Last but not the least, We owe to our family including our parents for their unconditional love and immense emotional support.

Publication List

[Optional] The main contributions of this research are either published or accepted or in preparation in journals and conferences as mentioned in the following list:

Journal Articles

- 1.

Conference Papers

- 1.

Additional Publications

Following is the list of relevant publications published in the course of the research that is not included in the thesis:

- 1.

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List of Abbreviations

IMPORTANT: Maintain alphabetical order (A–Z) when adding new abbreviations

AI	=	Artificial Intelligence
API	=	Application Programming Interface
CSE	=	Computer Science & Engineering
DBMS	=	Database Management System
IoT	=	Internet of Things
ML	=	Machine Learning

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Abstract

The abstract should contain a summary of the work presented in this report in a single paragraph. The paragraph should provide a general background of the problem, methodology and results.

Chapter 1

Introduction

[Must be present in FYDP-1 Report and also in Final Report]

Every chapter should start with 1-2 sentences on the outline of the chapter.

1.1 Thesis/Project Overview

This section should clearly present the background and a problem statement that your project aims to solve. The Thesis/Project Overview section provides a concise introduction to the entire work, giving readers a quick understanding of what the project is, why it is important, and how it is carried out. It typically includes the problem statement, objectives, scope, methodology, and expected outcomes. The overview should briefly describe the context of the problem, the motivation behind choosing it, and the proposed solution approach, without going into detailed technical explanations.

In a thesis (research) work, for example, an overview of an AI-based disease prediction system may mention the need for accurate diagnosis, the use of machine learning models, dataset sources, and expected improvement in prediction accuracy. In a project (system/software) work, such as a web-based student management system, the overview may highlight the need for automation, key features (student records, result management), technologies used (e.g., web framework, database), and expected benefits like efficiency and accuracy. Overall, this section acts as a snapshot of the entire work, helping readers quickly grasp the purpose, approach, and significance of the project.

1.2 Motivation

The Thesis/Project Motivation section explains why the problem was chosen and what inspired the work. It highlights the real-world need, existing challenges, and importance of solving the problem, linking them to academic, industrial, or societal relevance. This section typically includes the background context, observed issues in current systems, and the potential impact of an improved solution, helping readers understand the driving force behind the project.

In a thesis (research) work, for example, the motivation for an AI-based disease prediction system may come from the need for early and accurate diagnosis, limitations of existing models, and the growing importance of AI in healthcare. In a project (system/software) work, such as a student management system, the motivation may arise from manual record-keeping inefficiencies, data errors, and time-consuming administrative processes, encouraging the development of an automated solution. Overall, this section clearly communicates the purpose and significance of the work, showing why it is worth undertaking. The computational motivation that encourages you to solve the problem should be clearly stated here. In addition, you can mention why solving this problem will benefit you.

1.3 Objectives

The Thesis/Project Objectives section clearly states the specific goals that the work aims to achieve, usually in concise, measurable points. It translates the motivation into actionable targets, outlining what the system or research will accomplish. Objectives should be clear, achievable, and aligned with the problem statement, often divided into a general objective (overall goal) and several specific objectives (step-by-step aims such as design, implementation, and evaluation).

In a thesis (research) work, for example, an AI-based disease prediction study may have objectives like: to develop an accurate prediction model, to evaluate performance using metrics such as accuracy and F1-score, and to compare results with existing methods. In a project (system/software) work, such as a web-based student management system, objectives may include: to design a user-friendly interface, to implement modules for student records and result management, and to ensure efficient and secure data handling. Overall, this section defines what the project intends to achieve and provides a roadmap for the entire work.

Enumerate the objectives in clear and specific terms.

1. *Objective 1*

2. *Objective 2*

3. *Objective 3*

1.4 Methodology

Here you put a brief summary of the methodology.

1.5 Thesis/Project Outcome

What are or could be the possible outcomes of your work? The outcomes achieved in this theiss/project are as follows:

- 1. Outcome 1 corresponding to Objective 1.*
- 2. Outcome 2 corresponding to Objective 2.*
- 3. Outcome 3 corresponding to Objective 3.*

1.6 Organization of the Report

Here put a chapter wise structure of the report in narrative form.

Chapter 2

Background

[Must be present in FYDP-1 Report and also in Final Report]

Every chapter should start with 1-2 sentences on the outline of the chapter.

2.1 Preliminaries

In this section, you have to provide the necessary background knowledge to understand the rest of the report [1].

2.2 Literature Review

The Literature Review section summarizes and analyzes existing research, systems, and technologies related to the project, showing what has already been done and where your work fits in. It typically includes a review of relevant papers, methods, tools, and findings, along with a brief comparison of their strengths and limitations. The goal is not just to describe past work, but to critically analyze it and identify trends, gaps, and opportunities that justify your proposed solution.

In a thesis (research) work, for example, a student working on an AI-based disease prediction model may review several studies using CNN, SVM, or hybrid models, compare their accuracy (e.g., 80–90%), and discuss limitations such as small datasets or lack of generalization. In a project (system/software) work, such as a web-based management system, the review may include existing applications or frameworks, discussing features, usability, and shortcomings like lack of scalability or poor user interface. Overall, this section demonstrates understanding of the domain, awareness of prior work, and the need for the proposed solution.

This section will contain your literature review.

2.2.1 Similar Applications

The Similar Applications section describes existing real-world systems, software, or applications that are closely related to your project. It focuses on practical implementations

(not just research papers) and explains their features, technologies, strengths, and limitations. The purpose is to show awareness of current solutions and to highlight how your system is different or improved.

In a thesis (research) work, for example, if the topic is AI-based disease prediction, this section may discuss existing applications like hospital decision-support systems or diagnostic apps, describing their capabilities and limitations (e.g., limited accuracy or lack of local data). In a project (system/software) work, such as a student management system, similar applications may include existing university portals or commercial software, analyzing features like record management, reporting, and user interface, while noting gaps such as poor usability or lack of customization. Overall, this section demonstrates practical relevance, comparison with real systems, and justification for the proposed solution.

2.2.2 Related Research

The Related Research section reviews and discusses previous academic studies, journal papers, and conference works that are directly connected to your thesis or project topic. It focuses on research methods, models, algorithms, datasets, and key findings, showing how other researchers have approached similar problems. The section should not only summarize these works but also compare them, highlight their strengths and limitations, and identify research gaps that your work aims to address.

In a thesis (research) work, for example, a student working on an AI-based disease prediction system may analyze several studies using techniques like CNN, SVM, or hybrid models, comparing their performance (e.g., accuracy, PR-AUC) and noting limitations such as small datasets or lack of generalization. In a project (system/software) work, such as a smart traffic system, the section may include research on traffic optimization algorithms or IoT-based solutions, discussing their effectiveness and constraints. Overall, this section demonstrates deep understanding of the research landscape and positions your work within existing scientific contributions.

2.3 Gap Analysis

Gap analysis identifies the limitations, shortcomings, or unexplored areas in existing research or systems and explains how the proposed work aims to address those gaps. Recommended Table Format. The table columns will include: Existing Work, Limitations, Identified Gap, and Proposed Solution. Here summarise the gap where you intend to work.

2.4 Summary

Chapter 3

System Architecture

[Must be present in FYDP-1 Report and also in Final Report]

Every chapter should start with 1-2 sentences on the outline of the chapter.

3.1 Requirement Analysis

3.1.1 Functional and Nonfunctional Requirements

The Functional and Non-Functional Requirements Analysis section defines what the system will do and how well it will perform, forming the foundation for design and implementation. Functional requirements describe the core features and operations of the system (e.g., user registration, data processing, report generation), while non-functional requirements specify quality attributes such as performance, security, usability, reliability, and scalability. This section should present requirements clearly—often using bullet points or tables—and may include use cases or scenarios.

In a thesis (research) work, for example, an AI-based prediction system may include functional requirements like data input, model training, and prediction output, and non-functional requirements like high accuracy, low computation time, and secure data handling. In a project (system/software) work, such as a web-based management system, functional requirements may include login, data entry, and report generation, while non-functional requirements may include fast response time (e.g., <2 seconds), user-friendly interface, and support for multiple users. Overall, this section ensures that the system is clearly defined, measurable, and aligned with user and system expectations.

3.1.2 Context Diagram

The Context Diagram section provides a high-level graphical overview of the system, showing how the proposed project interacts with external entities (users, systems, or organizations) through data flows. It represents the entire system as a single process (black box) and illustrates only the inputs and outputs, without showing internal details. The section should include a labeled diagram along with a short explanation of each

external entity and the type of data exchanged.

For example, in a thesis or project work like a Student Management System, the context diagram would show the system at the center, with external entities such as Student, Teacher, and Admin. Data flows may include login requests, student information, marks entry, and result reports. The accompanying description should briefly explain how each entity interacts with the system. Overall, this section helps readers quickly understand the system boundary, scope, and interactions, serving as a foundation for more detailed design diagrams.

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3.1.3 UI Design

if required (may be for project work)

3.2 Detailed Methodology and Design

The Detailed Methodology and Design section explains how the project is carried out and how the system is structured, translating objectives into a clear technical plan. It typically includes the overall approach (research method or development model), system architecture, algorithms or models used, data collection/preprocessing (if any), and design components such as modules, database schema, and flow diagrams (e.g., flowchart, DFD, UML). This section should describe the step-by-step process from input to output, along with tools, technologies, and design decisions.

In a thesis (research) work, for example, an AI-based system may describe dataset collection, preprocessing, feature extraction, model architecture (e.g., CNN-BiLSTM), training process, and evaluation strategy, often supported by block diagrams. In a project (system/software) work, such as a web-based application, it may include system architecture (frontend–backend–database), module design (login, data management), database tables, API flow, and user interface design. Overall, this section demonstrates a well-structured, logical, and implementable plan, showing how the proposed solution is designed and executed.

3.3 Tools and Technologies Used

The Tools and Technologies Used section lists and briefly explains the software, hardware, programming languages, frameworks, and platforms utilized to develop the thesis

or project. Its purpose is to show how the system was implemented and why specific tools were chosen, often mentioning their role in development, testing, or deployment. This section may be presented in a table or categorized list (e.g., Programming Language, Framework, Database, Hardware, Development Tools).

In a thesis (research) work, for example, an AI-based project may use Python, TensorFlow/PyTorch, NumPy, and Jupyter Notebook, along with tools for data preprocessing and visualization. In a project (system/software) work, such as a web-based application, the technologies may include HTML, CSS, JavaScript, a backend framework (e.g., Django or Node.js), and a database like MySQL, along with version control tools like Git. The section should also briefly justify choices—for instance, selecting Python for its rich AI libraries or a specific framework for scalability. Overall, this section demonstrates the technical foundation and implementation environment of the work.

3.4 Work Division and Team Management

The Work Division and Team Management section explains how responsibilities were distributed among team members and how the project was coordinated and managed. It typically describes each member's role (e.g., development, design, testing, documentation), the task allocation strategy, and how collaboration was maintained using meetings, version control, or project management tools. This section may include a table showing team members and their assigned tasks, along with a brief explanation of workflow, communication, and timeline management (e.g., milestones or Gantt chart).

For example, in a thesis/project team of three students developing a web-based system, one member may handle frontend design (UI/UX), another may work on backend development and database, and the third may focus on testing, integration, and documentation. Team management may involve weekly meetings, task tracking using tools like Git or Trello, and collaborative coding via version control. Overall, this section demonstrates organized teamwork, clear responsibility distribution, and effective coordination, ensuring the successful completion of the project.

In a two-column table, the left column will contain the names of the team members, and the right column will list the tasks of the project/thesis completed by the respective team members.

3.5 Time Frame

There will be a Gantt chart that illustrates the different phases of the project work on the total timeline, along with the specific roles of each team member in those phases.

3.6 Summary

Chapter 4

Implementation and Results

[Must be present in Final Report. Incomplete version might be included in FYDP-1 Report, however it is optional.]

Every chapter should start with 1-2 sentences on the outline of the chapter.

4.1 Environment Setup

4.2 Testing and Evaluation

The Testing and Evaluation section explains how a system is tested and how well it performs, using test cases, metrics, and experimental results to validate its effectiveness. In a thesis (research) work, this typically involves dataset-based evaluation and performance metrics—for example, an AI disease detection model may be tested on unseen data and achieve 91% accuracy, outperforming an existing model with 85%, with analysis explaining the improvement. In a project (software/system) work, it focuses more on functional and performance testing—for example, a web-based student management system may be tested with multiple users, showing correct outputs for all test cases and maintaining an average response time of 1.2 seconds under load. In both cases, the section provides measurable evidence, comparison, and analysis to prove that the system meets its objectives

4.3 Results and Discussion

The Results and Discussion section presents the key findings of the work and explains their meaning in relation to the project objectives. It includes a clear presentation of results (tables, graphs, outputs), followed by interpretation—why the results are good or not, what patterns are observed, and how they compare with existing work. In a thesis (research) work, this often involves analyzing model performance and scientific outcomes—for example, an AI model achieving 92% accuracy may be discussed in terms of improved feature extraction, comparison with earlier studies (e.g., 85%), and reasons for any limitations such as overfitting or dataset bias. In a project (software/system) work, it focuses on sys-

tem behavior and usability—for example, a web application showing stable performance with 1.2-second response time under 100 users may be discussed in terms of efficiency, scalability, and user experience, along with any constraints like performance degradation at higher loads. Overall, this section not only reports results but critically interprets them, highlights strengths and weaknesses, and connects outcomes to the original objectives

4.4 Summary

Chapter 5

Standards and Design Constraints

[Must be present in FYDP-1 Report and also in Final Report]

Every chapter should start with 1-2 sentences on the outline of the chapter.

5.1 Compliance with the Standards

Only mention the standards that are related to your project. This list is not complete. For each of the standards discuss the alternates with pros and cons and rationale of selection.

5.1.1 Software Standards

The Software Standards section describes the guidelines, protocols, and best practices followed during development to ensure the system is reliable, secure, maintainable, and consistent. It typically covers coding standards (naming conventions, formatting), documentation practices, version control, testing standards, security guidelines, and any relevant international or industry standards (e.g., ISO/IEEE). The purpose is to show that the project was developed in a systematic and professional manner, following accepted engineering practices.

For example, in a thesis/project work, a student may follow coding standards like proper indentation and meaningful variable names, use Git for version control, apply testing standards such as unit and integration testing, and ensure security practices like password hashing and input validation. In a web-based system, they may follow REST API standards and basic OWASP security guidelines to prevent vulnerabilities. Overall, this section demonstrates that the software is developed with quality assurance, consistency, and adherence to recognized standards, making it robust and maintainable.

5.1.2 Hardware Standards

The Hardware Standards section describes the hardware specifications, guidelines, and best practices followed to ensure the system is safe, reliable, compatible, and efficient. It typically includes details such as required hardware components (e.g., microcontrollers,

sensors, servers, PCs), configuration specifications (CPU, RAM, storage), power requirements, interfacing standards, and compliance with safety or industry standards. The purpose is to show that the hardware used is appropriate, standardized, and suitable for the system's operation.

For example, in a thesis/project work involving an IoT-based system, the student may use ESP32/Arduino boards, temperature sensors, and regulated power supply, ensuring proper voltage levels (e.g., 5V/3.3V), secure wiring, and standard communication protocols like I2C, SPI, or UART. In a software-only project, this section may describe the minimum system requirements such as a computer with specific CPU, RAM, and OS configuration for deployment. Overall, this section demonstrates that the hardware setup follows standard practices for performance, safety, and compatibility, supporting reliable system operation.

5.1.3 Communication Standards

The Communication Standards section describes the protocols, data formats, and rules used for data exchange between different components of the system (e.g., client-server, device-device, or module-module). It explains how reliable, secure, and consistent communication is ensured, including aspects like network protocols, API design, message formats, error handling, and security measures. The goal is to show that the system follows standardized and interoperable communication practices.

For example, in a thesis/project work, a web-based application may use HTTP/HTTPS protocols with RESTful APIs, exchanging data in JSON format, and applying authentication tokens (e.g., JWT) for secure communication. In an IoT-based project, communication may follow protocols like MQTT, HTTP, or WebSocket, with devices sending sensor data to a cloud server. The section may also mention error handling, data validation, and encryption to ensure reliability and security. Overall, this section demonstrates that the system uses standard, efficient, and secure communication methods for proper interaction between its components.

5.2 Constraints and Limitations

The Constraints and Limitations section explains the practical restrictions faced during the project and the inherent weaknesses of the proposed solution. It typically covers constraints such as limited time, budget, hardware resources, dataset size, tools, or technical expertise, as well as system limitations like reduced accuracy, scalability issues, dependency on internet or specific platforms, or restricted functionality. This section should be honest and realistic, showing awareness of what the system cannot fully achieve and why.

For example, in a thesis (research) work, an AI model may be limited by a small or imbalanced dataset, leading to lower generalization performance despite good training accuracy. In a project (system/software) work, such as a web application, limitations

may include performance degradation under high user load or dependency on stable internet connectivity. Overall, this section demonstrates critical evaluation and transparency, helping readers understand the scope, challenges, and areas for future improvement.

5.2.1 Economic Constraint

The Economic Constraints section explains the financial limitations and affordability considerations that influence the design, development, and deployment of the system. It identifies cost-related factors such as budget limits, hardware/software expenses, maintenance cost, and scalability cost, and describes how the project is designed to remain cost-effective. In a thesis (research) work, for example, a student may use open-source tools (e.g., Python, TensorFlow) and publicly available datasets to minimize expenses while still achieving strong results. In a project (system/software) work, such as a web-based management system, the design may focus on low-cost hosting, minimal hardware requirements, and efficient resource usage—for instance, developing a system that runs on standard servers instead of expensive infrastructure. Overall, this section demonstrates that the solution is economically feasible and sustainable, especially within real-world budget constraints

5.2.2 Environmental Constraint

The Environmental Constraints section describes how the system is affected by environmental factors and how it impacts the environment, including issues like energy consumption, hardware usage, heat generation, e-waste, and sustainability. It also explains any limitations due to infrastructure (e.g., power availability, network conditions) and how the design addresses them. In a thesis (research) work, this might involve analyzing computational cost—for example, an AI model may require high GPU power, so the study can discuss optimizing the model to reduce energy consumption or using efficient architectures. In a project (system/software) work, it may focus on deployment—for example, a web system designed to run on low-power servers to reduce electricity usage.

Green computing can be incorporated by adopting environmentally friendly practices such as using energy-efficient algorithms, cloud-based shared resources instead of dedicated hardware, low-power devices, code optimization to reduce CPU usage, and minimizing unnecessary data processing or storage. For instance, a student project could implement a lightweight machine learning model that reduces computation time by 30%, thereby lowering power consumption and contributing to sustainable computing. Overall, this section shows awareness of environmental impact and demonstrates efforts to design a more sustainable and eco-friendly system. To estimate the environmental impact of computation, a simple electricity-based carbon model is applied. Using the Bangladesh grid emission factor (≈ 0.67 kg CO₂ per kWh), the carbon footprint is computed from the total energy used during model training. A typical large deep-learning workflow may require a 250 W GPU for multiple long training runs, resulting in significantly higher energy use and emis-

sions. In contrast, the proposed LitePhospho model uses a lightweight architecture that trains on a 100 W CPU/small GPU for fewer hours. As a result, the total computational carbon emission is considerably lower.

5.2.3 Ethical Constraint

The Ethical Constraints section explains how the project adheres to moral principles and professional responsibilities during development and deployment. It addresses issues such as data privacy, confidentiality, informed consent, fairness, transparency, avoidance of bias, and responsible use of technology. The section identifies potential ethical risks—like misuse of user data, biased AI decisions, or plagiarism—and describes the measures taken to prevent them. In a thesis (research) work, for example, an AI-based system using personal or medical data must ensure data anonymization, secure storage, and proper consent from data sources, while also evaluating and reducing algorithmic bias. In a project (system/software) work, such as a web or mobile application, ethical considerations may include secure authentication, protection of user information, and clear user policies. Overall, this section demonstrates that the system is developed in a way that is fair, responsible, and aligned with professional ethical standards.

5.2.4 Health and Safety Constraint

The Health and Safety Constraints section explains how the project considers and minimizes risks to users, developers, and the environment during development and operation. It identifies potential hazards—such as electrical risks (in hardware/IoT projects), prolonged screen exposure, poor ergonomics, data privacy concerns, or system misuse—and describes measures taken to ensure safety and compliance with standards. In a thesis (research) work, for example, an AI-based healthcare system should ensure data privacy, secure handling of sensitive information, and avoidance of misleading outputs that could affect patient decisions. In a project (system/software) work, such as an IoT-based smart device, safety measures may include proper circuit insulation, low-voltage design, secure connections, and user guidelines to prevent accidents. For software-only systems, considerations may include reducing eye strain through UI design, preventing data loss, and ensuring secure authentication. Overall, this section demonstrates that the system is designed with user well-being, safety standards, and risk mitigation in mind.

5.2.5 Social Constraint

The Social Constraints section explains how the project interacts with and is influenced by societal factors, and how it affects users and communities. It considers issues such as accessibility, cultural sensitivity, digital divide, user acceptance, ethical use, and inclusiveness. The section identifies potential social limitations—like lack of internet access, low digital literacy, language barriers, or resistance to new technology—and describes how the system design addresses them. In a thesis (research) work, for example, an AI-based

application may consider bias in data and ensure fairness so that different groups are not disadvantaged. In a project (system/software) work, such as a mobile app, the system may include local language support (e.g., Bangla), simple user interface, and low-bandwidth operation to make it usable for a wider population. Overall, this section demonstrates awareness of the system’s social impact and responsibility, ensuring it is usable, fair, and beneficial to society.

5.2.6 Sustainability

The Sustainability section explains how the project is designed to be environmentally, economically, and socially sustainable over the long term. It highlights practices that reduce resource consumption, ensure efficient operation, and maintain usability for future users. This includes aspects such as energy efficiency, scalability, maintainability, use of open-source technologies, and minimal environmental impact. In a thesis (research) work, for example, a student may develop an optimized AI model that achieves high accuracy with reduced computational cost, thereby lowering energy consumption. In a project (system/software) work, such as a web or IoT system, sustainability may involve low-power hardware, efficient code, cloud-based shared resources, and easy system updates to extend system lifespan. Overall, this section demonstrates that the solution is not only effective now but also viable, efficient, and responsible for long-term use and future development.

5.2.7 Lifelong Learning and Personal Development

The Lifelong Learning and Personal Development section highlights how the thesis or project contributed to the student’s continuous learning ability and overall skill development. It reflects on the knowledge and competencies gained during the work, such as learning new programming languages, tools, frameworks, or research methodologies, as well as improving problem-solving, critical thinking, teamwork, and communication skills. In a thesis (research) work, for example, a student may learn how to read and analyze research papers, design experiments, and implement advanced models (e.g., deep learning), which prepares them for future research or higher studies. In a project (system/software) work, a student might gain hands-on experience in system design, debugging, version control, and user interaction—for instance, developing a web application may enhance full-stack development skills and real-world problem-solving ability. Overall, this section demonstrates how the work has strengthened the student’s ability to adapt, learn independently, and grow professionally beyond the project itself.

5.3 Cost Analysis

The cost-benefit analysis will assess whether the proposed system is economically and practically worthwhile by comparing the total cost of development and operation with the expected benefits. It typically includes cost components such as hardware, software, de-

velopment time, maintenance, and training, and benefits such as time savings, improved accuracy, efficiency, scalability, or social impact. In a thesis (research) work, this may focus on research value—for example, an AI model developed using open-source tools may have minimal financial cost but high benefit in terms of improved prediction accuracy and contribution to knowledge. In a project (system/software) work, it is often more practical—for example, a web-based management system costing BDT 50,000 to develop may reduce manual work by 70%, save administrative time, and improve data accuracy, making it cost-effective in the long run. Overall, this section demonstrates that the proposed solution provides greater benefits compared to its cost, justifying its implementation. Provide a cost analysis in terms of budget required and revenue model. In case of budget, you must show an alternate budget and rationales.

5.4 Complex Engineering Problem

5.4.1 Complex Problem Solving

The Complex Engineering Problem (CEP) section explains why a thesis or project involves significant engineering difficulty, requiring analysis, judgment, and integration of multiple factors rather than a routine solution. To justify CEP (as per BAETE/Washington Accord style), you should explicitly show that the work satisfies WP1, WP2, and WP3.

WP1 (Depth of Knowledge Required): The problem demands advanced and specialized knowledge beyond basic coursework.

WP2 (Range of Conflicting Requirements): The solution must balance multiple, often conflicting constraints (e.g., accuracy vs. cost, performance vs. energy, usability vs. security).

WP3 (Depth of Analysis Required): The problem requires in-depth analysis, modeling, experimentation, and evaluation of alternatives, not a straightforward implementation.

Example (Thesis Work – AI-based Disease Prediction)

This work can be justified as a CEP because:

WP1: It requires knowledge of machine learning, deep learning architectures, and data preprocessing techniques.

WP2: It must balance high accuracy with limited dataset size, computational cost, and ethical concerns like data privacy.

WP3: It involves extensive experimentation (model tuning, comparison with baseline models, performance evaluation using metrics like accuracy and F1-score).

Example (Project Work – Smart Traffic Management System)

WP1: Requires knowledge of IoT, networking, and real-time systems.

WP2: Must handle conflicts between real-time performance, cost of sensors, network latency, and scalability.

WP3: Requires analysis of traffic patterns, algorithm design for signal optimization, and testing under different traffic scenarios.

Overall, the CEP section should clearly demonstrate that the work involves advanced knowledge, multiple constraints, and rigorous analysis, thereby qualifying it as a Complex Engineering Problem rather than a simple or routine task.

In this section, provide a mapping with problem solving categories. For each mapping add subsections to put rationale (Use Table 5.1). For P1, you need to put another mapping with Knowledge profile and rational thereof.

Table 5.1: Mapping with complex problem solving.

WP1 Dept of Knowl- edge	WP2 Range of Con- flicting Require- ments	WP3 Depth of Analysis	WP4 Familiarity of Issues	WP5 Extent of Applicable Codes	WP6 Extent of Stake- holder Involve- ment	WP7 Inter- dependence
✓	✓	✓				

5.4.2 Engineering Activities

The Complex Engineering Activities (CEA) section explains how the student has carried out non-routine, technically demanding engineering tasks that involve design, implementation, and decision-making under constraints. It demonstrates that the work is not just theoretical but includes practical realization, integration of components, testing, and iterative improvement. To justify CEA (aligned with BAETE/Washington Accord), you should explicitly show that the work satisfies EA1 and EA2:

EA1 (Range of Resources and Techniques): The project uses diverse tools, technologies, and methods, such as programming frameworks, hardware/software integration, simulations, datasets, and testing tools.

EA2 (Level of Interaction and Integration): The work involves integration of multiple subsystems or components, requiring coordination between different modules, technologies, or stakeholders.

Example (Thesis Work – AI-based Disease Detection)

EA1: Uses Python, TensorFlow/PyTorch, data preprocessing techniques, and performance evaluation tools.

EA2: Integrates data collection, preprocessing pipeline, model training, evaluation, and possibly a user interface for prediction, forming a complete system.

Example (Project Work – Smart Home Automation System) EA1: Uses microcontrollers (e.g., Arduino/ESP32), sensors, mobile app development, and networking protocols.

EA2: Integrates hardware (sensors/actuators), software (control logic), and communication (Wi-Fi/Bluetooth) into a unified system.

Overall, this section should clearly show that the student has engaged in hands-on engineering activities involving multiple tools and system integration, proving the work qualifies as Complex Engineering Activities, not just a simple or isolated task.

In this section, provide a mapping with engineering activities. For each mapping add subsections to put rationale (Use Table 5.2).

Table 5.2: Mapping with complex engineering activities.

EA1 Range of re- sources	EA2 Level of Interac- tion	EA3 Innovation	EA4 Consequences for society and environment	EA5 Familiarity
✓	✓			

5.5 Summary

Chapter 6

Conclusion

[Must be present in FYDP-1 Report and also in Final Report. Might be incomplete in FYDP-1 Report.]

Every chapter should start with 1-2 sentences on the outline of the chapter.

6.1 Summary

6.2 Limitation

6.3 Future Work

References

- [1] Jon Kleinberg and Eva Tardos. *Algorithm design*. Pearson Education India, 2006.